

1. What problem do traditional Recurrent Neural Networks face when learning long sequences, and how do Long Short-Term Memories solve it?

Answer conceptually without using mathematical formulas.

2. Explain the difference between cell state () and hidden state () in an LSTM. Why are both needed?

Answer conceptually without using mathematical formulas.

3. What is the role of the Forget Gate () in an LSTM? What problems could happen if it did not exist?

Answer conceptually without using mathematical formulas.

4. How does the Input Gate () help an LSTM decide what new information should be stored in memory?

Answer conceptually without using mathematical formulas.

5. What is the purpose of the Output Gate () and how does it help create the hidden state ()?

Answer conceptually without using mathematical formulas.